WASTEWISE SCAN · PRODUCT REQUIREMENTS AND SPECIFICATIONS

WasteWise Scan product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-24	Amish Chadha	Kit 1.8 upgrade: WSC-REQ-001 becomes the product requirements and specifications WSC-PRS-001; R1 to R14 migrated to PR-001 to PR-014 with the same targets, verification and status, each traced to a user requirement in WSC-URS-001, with tolerance and units; PR-015 (cell safety), PR-016 (buildable) and PR-017 (optical crosstalk, fix F1) added; statuses after fixes F1 to F3 of WSC-DEC-001 (0.64 s, 21.9 h, 239 g and 259 g with cap)
0.7	2026-10-02	Amish Chadha	R6 restated for 2.0 mm shell walls (238 g, 258 g with cap); R1 note carries the paper band study; no status changed
0.6	2026-10-02	Amish Chadha	R6 and R13 status notes record Amish's decisions of 2026-10-02 (WSC-DEC-001); no status changed
0.5	2026-10-02	Amish Chadha	Status from WSC-CAL-001 v0.3 for the constructable design (WSC-DDR-004);

			R6 not met; R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R4, R6, R11 and R12 restated per WSC-DDR-003; status from WSC-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 status from WSC-CAL-001; R12 redefined per WSC-DDR-001 D2; R13 points to the draft record in WSC-DDR-002
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

WasteWise Scan product requirements and specifications

What the scanner must do, each requirement traced to a user need in WSC-URS-001 (links in the generated traceability matrix WSC-RTM-001). This document replaces the requirements WSC-REQ-001 (kit 1.8, STANDARDS section 22): each old requirement R1 to R14 is now PR-001 to PR-014 with the same target, verification and status, and the old number in "Was". Status is from the calculation note WSC-CAL-001 (tags in brackets) for option A, the current design. **On paper, 8 of the 17 are met on paper or by design, PR-006 (mass) is not met (259 g with the cap against 250 g), PR-001, PR-007 and PR-013 are at risk, PR-003, PR-010, PR-015 and PR-017 are to verify at TRL 4, and PR-012 is over the value-engineering target.** Three fixes made in this revision under Amish's standing instruction of 2026-10-23 (WSC-DEC-001 F1 to F3) changed PR-002, PR-004, PR-005, PR-006, PR-010, PR-015 and PR-017. Targets remain proposals for review with the co-design partner (first candidate SWaCH, Pune; WSC-DEC-001).

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Identify common resins	The scanner shall name PET, HDPE, PP, PS and PVC, plus "other" and "unknown", on clean, dry, non-black items of 1 mm wall or more	95	min	% correct	Analysis, paper band study WSC-CAL-001 B4 to B7 (TRL 3); labelled item set of 200 or more (TRL 4)	At risk: 89.5 % with eight bands at 1.5 % band scatter, 96.7 % at 1.0 %; PE against PP 93 %; 1,650 nm band centroid 1,628 nm [B2, B4 to B7]	R1
PR-002	UR-002	Performance	Fast result	Button press to result shall take no longer than the target	1.5	max	s	Analysis, timing budget WSC-CAL-001 E (TRL 3); timed at TRL 4	Met on paper: 0.64 s with the four-cycle sequence of fix F2 [E2]	R2
PR-003	UR-003	Software	Fail safe on unreadable items	Black, dark, wet or very thin items shall read "unknown"; a wrong resin shall be shown for no more than the target share of all scans	2	max	% of scans	Analysis of the black threshold WSC-CAL-001 C6 (TRL 3); confidence threshold study on items (TRL 4)	To verify at TRL 4: black items fall well below the 0.08 reflectance threshold [C6]; the wrong-result rate needs item data	R3
PR-004	UR-004	Performance	Work in sunlight	With the shroud pressed to the item, the result in direct sun up to 100 klx shall match the indoor result; clear and translucent	0.5	max	% of signal, ambient error	Analysis, stray light WSC-CAL-001 D (TRL 3); indoor and sun test (TRL 4)	Met on paper (estimate): 0.08 % on opaque items, 0.21 % on backed clear items with fix F2; unbacked clear items	R4

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				items are backed with the calibration cap, and the scanner shall refuse a result ("shade or back with cap") when its dark level shows light through an unbacked item					refused [D5 to D7]; rests on the assumed cap leak and ambient change (STEM S5)	
PR-005	UR-005	Electrical	Last a shift	The scanner shall run a shift at 4 scans per minute on one charge and charge from USB-C	8	min	h	Analysis, power budget WSC-CAL-001 F (TRL 3); measured at TRL 4	Met on paper: 21.9 h [F2]	R5
PR-006	UR-006	Physical architecture	One-handed and light	Mass including the sun hood and the carried calibration cap shall not exceed the target; body no larger than 170 x 65 x 40 mm (6.7 x 2.6 x 1.6 in) plus shroud and hood	250	max	g	Analysis from the model volumes WSC-CAL-001 A (TRL 3); weighing at TRL 4 (WSC-DEC-001)	Not met: 239 g, 259 g with the cap (9 g over); body 160 x 62 x 34 mm [A1, A3]	R6
PR-007	UR-007	Usability	Readable by anyone	The result shall show as resin code, name, colour and icon, legible at arm's length in direct sun,	3	min	contrast ratio	Analysis, contrast WSC-CAL-001 I (TRL 3); design review with users	At risk: about 5.2:1 where the sun hood shades the screen; about 1.6:1 with the	R7

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				in the local language, with no reading needed to use					sun near the screen normal or over the open side [I3, I4]	
PR-008	UR-008	Software	Local price grade	The scanner shall show a price grade from a table the user or cooperative edits on the device, offline	1	min	editable local table	Inspection of the design (TRL 3); design review with the cooperative	Met by design	R8
PR-009	UR-009, UR-017	Software	Log scans	The scanner shall store scans on the device with time, result, confidence and spectrum and export them as CSV over USB, with no cloud needed	10000	min	scans	Analysis, storage WSC-CAL-001 G (TRL 3)	Met on paper: about 131,000 scans in an 8 MB partition [G1]	R9
PR-010	UR-010	Physical architecture	Rugged	The scanner shall survive a drop onto concrete from the target height, meet IP54 and work from 0 to 45 C	1.2	min	m drop	Analysis of drop loads WSC-CAL-001 J (TRL 3); drop and spray tests (TRL 4)	To verify at TRL 4: corner drop about 1,200 g mean on an assumed 1 mm crush; the 11 x 10 mm cell strap (fix F3) carries the 869 N peak with a margin of 1.45 [J1, J2, J4]	R10
PR-011	UR-011	Usability	Stay calibrated	A white reference	10	max	s	Inspection of the design	Met by design: 5 %	R11

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				check and an open-air crosstalk reading with the calibration cap shall take no longer than the target, with a warning when a check is overdue or drift exceeds 5 %				and WSC-CAL-001 J3 (TRL 3)	drift after about 12.5 C of change [J3]	
PR-012	UR-012	Physical architecture	Low cost	Parts cost per unit against the value-engineering target, with no custom PCB for the first build	150	target	USD	Priced BOM, WSC-CAL-001 K (TRL 3)	Over target: USD 167.50, USD 17.50 over the USD 150 value-engineering target [K2]	R12
PR-013	UR-013, UR-017	Software	Work with WasteWise-ml	Low-confidence scans shall hand off to WasteWise-ml; scan records shall use the shared, documented format of WSC-DDR-002	1	min	shared record	Inspection, interface note with WasteWise-ml	At risk: record decided on this side (WSC-DDR-002 v0.4), not yet adopted by WasteWise-ml	R13
PR-014	UR-014	Safety	Eye-safe illumination	LED emission shall stay in the IEC 62471 exempt group, the LEDs pulsing only during a scan, with a	100	max	W/m2 at 200 mm	Analysis WSC-CAL-001 H (TRL 3); datasheet values and measurement (TRL 4)	Met on paper (estimate): 3.9 W/m2 at 200 mm; retinal 3.8 % of the limit [H2, H3]	R14

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				hardware limit on on-time						
PR-015	UR-015	Safety	Safe cell	The scanner shall use one protected 18650 cell from a named maker in a cradle and strap that hold it against the peak corner-drop load, charged only by the board charger and not above 45 C	870	min	N cell retention (peak)	Analysis WSC-CAL-001 J2, J4 (TRL 3); STEM S10, S17, S18; drop test at TRL 4	To verify at TRL 4: the 11 x 10 mm strap (fix F3) carries 869 N at a bending margin of 1.45, on an assumed crush [J4]	
PR-016	UR-016	Physical architecture	Buildable by a local maker	The first prototype shall be built from bought modules, FDM-printed PETG and TPU parts, perfboard and screws, with through-hole soldering, hand tools and a drill	0	max	custom PCBs	Inspection of the build plan WSC-BLD-001 and the BOM (TRL 3)	Met by design (WSC-BLD-001; the 8.4 mm hole in the LED window of fix F1 needs a drill and an 8 mm diamond core drill)	
PR-017	UR-001, UR-011	Performance	Optical crosstalk	LED light reaching the detector without leaving through the item shall stay under the target share of the signal after the open-air	0.5	max	% of signal	Analysis WSC-CAL-001 C5, C7, C8 and STEM S12 (TRL 3); wall-opacity coupon, open-air and dirty-window readings (TRL 4)	To verify at TRL 4: 0.003 % with the baffle through the glass on assumed leak values (fix F1; 10 % before the fix); the	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				reading, with a 10 % change of the window state between readings					tube wall needs an opacity of 2 x 10^{-2} or better [C8]	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- "Clean and dry" means rinsed or wiped items without labels over the scan spot. Real waste is dirtier; the classifier must reject rather than guess.
- About 10 to 15 % of items in a mixed post-consumer stream are black or very dark (estimate; varies strongly by region and should be measured with users).
- The first prototype's accepted cost of about USD 164 covers one scanner with its cap and sun hood; the USD 150 value-engineering target assumes quantity prices for LEDs and the photodiode (WSC-DDR-001, D2), and is the price a cooperative pays for a shared scanner (WSC-DEC-001).
- Four scans per minute is a busy spot-check pace, not every item: most items are sorted by eye and the scanner settles doubtful ones.
- Resin prices vary by city and season; the device stores grades (for example A, B, C, reject) and a local price list, not global prices.
- PR-006 counts the calibration cap because it is carried for backing clear items (decided by Amish, 2026-10-02).